

Chapter 1

INTRODUCTION

With the rapid advancement of computer vision, deep learning, and 3D graphics technologies, there is a growing need for intelligent systems capable of automating the understanding and digital reconstruction of physical objects from images [1, 2]. Conventional approaches to 3D digitisation typically rely on manual modelling, specialised photogrammetry equipment, or controlled multi-view capture setups that require significant time, expertise, and infrastructure [3]. These limitations create bottlenecks for industries such as e-commerce, augmented reality (AR), robotics, and industrial inspection.

The Automated Single-Image to 3D Asset and Point Cloud Generation System is a fully integrated, end-to-end system designed to take any single input image and automatically produce a textured 3D mesh and a segmented, colour-labelled 3D point cloud of the primary object within it [4]. The pipeline eliminates the need for manual prompt engineering, controlled lighting, or multi-view capture by combining state-of-the-art foundation models into a unified automated workflow.

The pipeline is composed of two major subsystems. The first is a **Universal Object Detection** system built on three complementary foundation models: GroundingDINO for zero-shot open-vocabulary object detection, Florence-2 for image captioning and part-level detection, and SAM2.1 for precise instance segmentation [5, 6, 7]. The second subsystem leverages **Hunyuan3D-2**, a two-stage generative model developed by Tencent, which produces a watertight 3D mesh via multi-view diffusion and synthesises high-resolution PBR textures using appearance flow [8].

The complete workflow is complemented by three standalone Python utility scripts. Script 1 provides an interactive auto-rotating 3D mesh viewer built on Open3D. Script 2 converts the generated GLB mesh into a dense, uniformly sampled point cloud using Poisson-disk surface sampling. Script 3 performs multi-method point cloud segmentation using DBSCAN Euclidean clustering, RANSAC plane fitting, and voxel decomposition [9, 10]. The pipeline was validated end-to-end on a ceramic mug object, demonstrating successful part detection, high-quality 3D reconstruction, and accurate point cloud segmentation.

1.1 Literature Review

A literature survey was conducted on existing research works related to object detection, image segmentation, single-image 3D reconstruction, and point cloud processing. The survey helped identify limitations of traditional approaches and highlighted the importance of foundation model integration for automated 3D digitisation pipelines.

1. **Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection (2023)**

This work proposes a zero-shot open-vocabulary object detector by fusing a DINO-based visual encoder with a BERT text encoder through cross-modal attention. The model accepts natural-language text prompts and returns grounded bounding boxes without requiring class-specific training data. The approach demonstrates strong generalisation to novel object categories but shows sensitivity to image contrast and lighting conditions, motivating the multi-pass adaptive enhancement strategy used in this pipeline [5].

2. **Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks (2024)**

Florence-2 introduces a unified vision-language model capable of image captioning, object detection, and open-vocabulary region description using a single sequence-to-sequence formulation. The model eliminates the need for task-specific heads and enables automatic prompt generation from raw images. In this pipeline, Florence-2 is used both for auto-captioning (to generate GroundingDINO prompts) and for part-level detection (to identify object sub-regions) [6].

3. **SAM 2: Segment Anything in Images and Videos (2024)**

SAM2 extends the Segment Anything Model to video sequences while retaining state-of-the-art image segmentation performance. The hierarchical Hierarchical Encoder and lightweight mask decoder enable prompt-based pixel-precise instance segmentation from bounding box inputs. SAM2.1 is used in this pipeline to generate binary masks for each detected object part, producing clean object crops for 3D generation [7].

4. **Hunyuan3D-2: Scaling Diffusion Models for High Resolution Textured 3D Assets Generation (2025)**

This paper presents a two-stage 3D generative framework: a Diffusion Transformer that generates watertight 3D meshes from single images using flow-matching diffusion, followed by an appearance-flow texture synthesis pipeline that bakes high-fidelity PBR textures. The system achieves high geometric accuracy and texture fidelity from a single input photograph, reducing 3D asset creation time to approximately three minutes on a consumer-grade GPU [8].

5. **TripoSr: Fast 3D Object Reconstruction from a Single Image (2024)**

TripoSr demonstrates feed-forward single-image 3D reconstruction using a transformer-based architecture trained on large-scale 3D datasets. While it achieves faster inference than diffusion-based approaches, the output texture quality is more limited and the system lacks an integrated detection and background removal preprocessing stage, requiring manually prepared background-free inputs [11].

1.2 Motivation

Many industries and organisations face significant barriers to 3D content creation due to the high cost, complexity, and time requirements of traditional modelling and photogrammetry workflows [3]. A single high-quality 3D model can take a skilled artist several hours to produce manually, or require dozens of calibrated photographs and specialised reconstruction software in photogrammetry workflows.

Furthermore, existing detection pipelines frequently fail on challenging imaging conditions such as dark, low-contrast, greyscale, or cluttered images that are common in real-world product photography [1]. Most detection systems assume controlled lighting and high-contrast scenes, making them unreliable in practical deployments where input image quality cannot be guaranteed.

The motivation behind this project is to eliminate these barriers by providing a fully automated, single-image-to-3D workflow that is robust to diverse imaging conditions [4]. The growing demand for 3D content in AR/VR product visualisation at Pixrity Private Limited creates a strong industry-driven need for such a system. By integrating adaptive image enhancement, multi-model detection fallbacks, and state-of-the-art generative 3D reconstruction into a single cohesive pipeline, the system enables rapid, reliable, and high-quality 3D digitisation from any input photograph.

1.3 Objectives of the Project

The primary objectives of this internship project are:

- To design and implement a Universal Object Detection Pipeline combining GroundingDINO, Florence-2, and SAM2.1, capable of robust detection under diverse image conditions including dark, low-contrast, and greyscale images.
- To develop automated image captioning and prompt generation using Florence-2 to enable fully prompt-free operation on arbitrary input images without any manual user input.
- To integrate Hunyuan3D-2 for single-image to textured 3D mesh generation via background removal, RGBA crop extraction, and two-stage generative reconstruction.
- To implement an interactive GLB/OBJ 3D mesh visualiser with auto-rotating Open3D rendering for inspection of generated models.
- To build a mesh-to-point-cloud conversion utility using Poisson-disk surface sampling and surface normal estimation.
- To implement a multi-method point cloud segmentation tool supporting DBSCAN Euclidean clustering, RANSAC plane fitting, and voxel-based decomposition with colour-labelled cluster visualisation.

- To validate the complete pipeline end-to-end on a real-world ceramic mug object across all pipeline stages.

1.4 Problem Statement

Most existing 3D digitisation approaches are fragmented, requiring separate tools for detection, background removal, 3D generation, and point cloud analysis, with no unified automated workflow connecting these stages. Detection systems do not natively interface with 3D generation tools; generative 3D models require carefully preprocessed background-free RGBA inputs; and generated mesh assets lack downstream semantic analysis capabilities [3].

Additionally, conventional detection pipelines assume favourable imaging conditions and fail on the dark, low-contrast, or noisy images commonly encountered in real-world product photography. The absence of adaptive preprocessing and multi-model fallback strategies means that a single difficult image can halt the entire digitisation workflow.

This project addresses these challenges by designing a tightly integrated pipeline with robust error handling, adaptive CLAHE image enhancement, and multi-model fallbacks at every stage from image loading through detection, segmentation, 3D generation, and point cloud analysis enabling fully automated 3D digitisation from any single input photograph.

Chapter 2

REQUIREMENT ANALYSIS

1.1 Requirement Analysis

The requirement analysis for the Automated Single-Image to 3D Asset and Point Cloud Generation System identifies, analyses, and documents the functional and non-functional requirements necessary to achieve a fully automated, single-image-to-3D workflow. This phase is critical for understanding the practical capabilities of each pipeline component and ensuring that the system satisfies the technical and operational expectations of both the development team at Pixrity Private Limited and the end users of the 3D digitisation workflow.

The analysis begins with the study of the limitations and inefficiencies in existing 3D digitisation approaches, including the fragmentation between detection, segmentation, 3D generation, and point cloud processing tools; the lack of robustness to challenging imaging conditions; and the absence of automated prompt generation for open-vocabulary detection models. These limitations reduce productivity and restrict adoption of 3D digitisation in resource-constrained environments.

The requirement analysis further considers the need for adaptive image preprocessing, multi-model detection fallback strategies, efficient GPU memory management across multiple large foundation models, and seamless inter-stage data transfer (bounding boxes to segmentation masks, masks to RGBA crops, RGBA crops to 3D meshes, meshes to point clouds). Each of these data handoffs must be robust to failure modes at individual stages to ensure end-to-end pipeline reliability.

Non-functional requirements including inference speed, GPU VRAM utilisation, output file format compatibility, and cross-platform script portability are also identified and documented, as these directly impact the practical deployment of the pipeline within Pixrity's AR/VR development workflow.

1.2 System Model

The Automated Single-Image to 3D Asset and Point Cloud Generation System adopts a sequential stage-based architecture consisting of four macro-stages: Input and Enhancement, Universal Detection, 3D Generation, and Point Cloud Processing. Each stage produces a well-defined output that serves as the input to the next stage, ensuring modularity and fault isolation.

The **Input and Enhancement Stage** handles raw image loading in any common format (JPEG, PNG, WebP, BMP), computes image statistics including mean brightness and contrast standard deviation, and applies adaptive CLAHE enhancement when the image is classified as

dark or low-contrast. This stage ensures that all subsequent models receive appropriately conditioned input regardless of original image quality.

The **Universal Detection Stage** uses three foundation models in sequence. Florence-2 generates a natural-language caption of the input image, which is automatically converted into GroundingDINO text prompts using noun extraction. GroundingDINO performs zero-shot open-vocabulary whole-object detection across up to three passes with progressive threshold relaxation. Florence-2 then performs open-vocabulary part detection on the detected object region. Finally, SAM2.1 generates pixel-precise binary masks for each detected part.

The **3D Generation Stage** extracts the primary object RGBA crop using the SAM2.1 mask and background removal (rembg with ONNX GPU runtime). The cropped RGBA image is fed into Hunyuan3D-2 Stage 1 (flow-matching diffusion for shape generation) followed by Stage 2 (appearance-flow UV texture synthesis). The textured 3D mesh is exported as a GLB file.

The **Point Cloud Processing Stage** converts the GLB mesh into a dense, uniformly distributed point cloud using Poisson-disk surface sampling. Surface normals are estimated and consistently oriented. DBSCAN Euclidean clustering decomposes the point cloud into semantically meaningful geometric clusters, which are exported as a colour-labelled PLY file.

The following user roles interact with the pipeline:

- **Pipeline User:** Provides a single input image and receives the textured GLB mesh and segmented PLY point cloud as outputs. No manual prompt engineering or parameter setting is required.
- **Developer / Researcher:** Interacts with the pipeline via Jupyter/Colab notebooks or CLI script interfaces, with access to configurable parameters including DBSCAN epsilon, mesh rotation speed, point cloud target size, and texture resolution.
- **AR/VR Integration Engineer:** Consumes the generated GLB assets directly in Pixrity's WebXR and AR development workflows for 3D product visualisation and immersive experience prototyping.

1.3 Functional Requirements

The system must satisfy the following functional requirements:

- The system shall accept any standard image format (JPEG, PNG, WebP, BMP) as input and automatically load and convert it to the appropriate internal format for each pipeline stage.
- The system shall compute image brightness and contrast statistics and apply adaptive CLAHE enhancement when the image is classified as dark (mean brightness < 0.30) or low-contrast (standard deviation < 0.15).
- The system shall automatically generate a natural-language caption of the input image using Florence-2 and convert it into a GroundingDINO-compatible text prompt without any manual user input.

- The system shall perform zero-shot open-vocabulary object detection using GroundingDINO with a multi-pass strategy supporting up to three detection passes with progressive threshold relaxation (0.20, 0.15, 0.10).
- The system shall perform open-vocabulary part-level detection using Florence-2's open vocabulary detection task on the detected object region.
- The system shall generate pixel-precise binary segmentation masks for each detected part using SAM2.1 with box-prompted multi-mask output and automatic quality-score-based mask selection.
- The system shall remove the image background using rembg with ONNX GPU runtime and produce a clean RGBA crop of the primary detected object.
- The system shall generate a watertight untextured 3D mesh from the RGBA crop using Hunyuan3D-2 Stage 1 flow-matching diffusion in 30–75 inference steps.
- The system shall synthesise high-fidelity PBR textures on the generated mesh using Hunyuan3D-2 Stage 2 appearance-flow UV mapping at configurable resolution (512, 1024, or 2048 pixels).
- The system shall export the final textured 3D mesh as a GLB file compatible with standard 3D viewers and WebXR platforms.
- The system shall convert the GLB mesh to a dense point cloud using Poisson-disk surface sampling with configurable target point count.
- The system shall estimate and consistently orient surface normals for the generated point cloud.
- The system shall segment the point cloud using DBSCAN Euclidean clustering and export a colour-labelled PLY file with each cluster assigned a distinct colour.
- The system shall provide an interactive auto-rotating 3D mesh and point cloud viewer using Open3D.

1.4 Non-Functional Requirements

The system must satisfy the following non-functional requirements:

Performance: The complete pipeline from input image to textured GLB mesh shall complete within 4 minutes on an NVIDIA T4 GPU (16 GB VRAM). Point cloud conversion and segmentation shall complete within 15 seconds on CPU.

Memory Management: The pipeline shall manage GPU VRAM efficiently through sequential model loading and unloading, using cuda empty cache between stages to prevent out-of-memory errors on a 16 GB VRAM GPU.

Robustness: The pipeline shall handle dark (mean brightness < 0.30), low-contrast, greyscale, and standard colour images through adaptive preprocessing and multi-pass detection fallbacks. The system shall not crash on first-pass detection failure but shall attempt fallback strategies automatically.

Output Compatibility: Generated GLB files shall be compatible with standard 3D viewers (Blender, Three.js, WebXR). PLY point cloud files shall be compatible with Open3D, MeshLab, and CloudCompare.

Scalability: The utility scripts shall support configurable parameters (point cloud size, texture resolution, DBSCAN parameters) to accommodate objects of varying complexity and detail levels.

Usability: The pipeline shall operate in a fully prompt-free mode requiring no manual text input from the user. All three utility scripts shall provide clean CLI interfaces via argparse for integration into automated batch processing workflows.

Hardware Requirements – Server/Compute Side:

- GPU: NVIDIA T4 or equivalent with minimum 16 GB VRAM
- CUDA: Version 12.1 or later
- RAM: Minimum 12 GB system RAM
- Storage: Minimum 50 GB for model weights and pipeline outputs
- Network: High-speed internet for initial HuggingFace Hub model downloads (approximately 12 GB total)

Software Requirements:

- Operating System: Linux (Ubuntu 20.04+) or Google Colab environment
- Python: Version 3.10 or later
- Deep Learning: PyTorch 2.x with CUDA 12.1
- Model Hub: HuggingFace Transformers (latest)
- 3D Processing: Open3D ≥ 0.16 , trimesh (latest)
- Image Processing: Pillow, OpenCV, NumPy, Matplotlib
- Background Removal: rembg with ONNX GPU runtime
- Development: Git, argparse, Jupyter/Colab Notebooks

Chapter 3

SYSTEM DESIGN

3.1 Architecture of the System

The Automated Single-Image to 3D Asset and Point Cloud Generation System uses a sequential stage-based architecture that guarantees modularity, fault isolation, and efficient data flow between components. The architecture is organised into four macro-stages connected by well-defined inter-stage data interfaces. Figure 3.1 illustrates the complete system architecture.

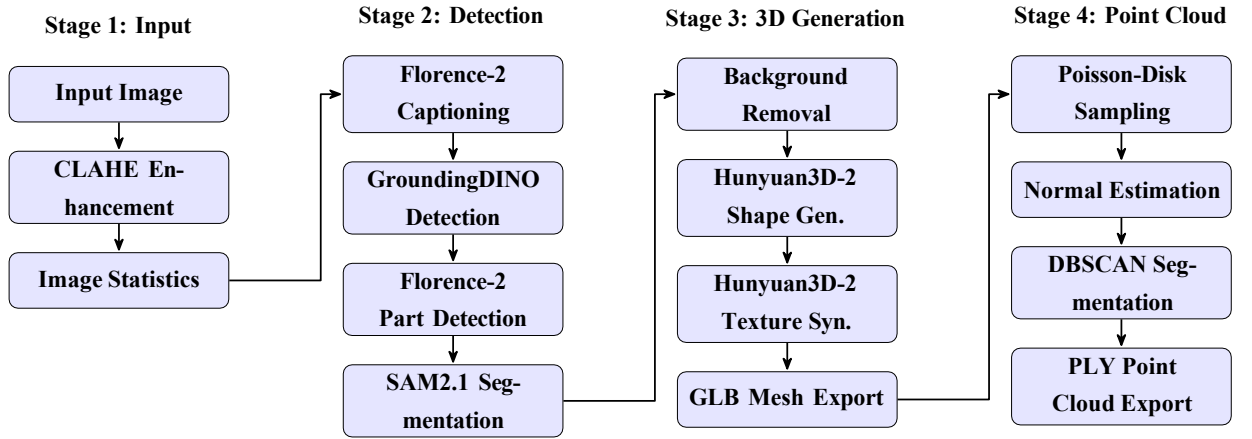


Figure 3.1: Architecture of the Automated Single-Image to 3D Asset and Point Cloud Generation System showing four macro-stages and data flow between components

The **Input and Enhancement Stage** is the entry point for all images entering the pipeline. The system computes mean brightness and contrast standard deviation of the input image and classifies it as dark, low-contrast, or standard. For dark or low-contrast images, adaptive CLAHE (Contrast Limited Adaptive Histogram Equalisation) is applied using OpenCV in the LAB colour space to improve local contrast without over-amplifying noise. The enhanced image is passed to the detection stage alongside the original, allowing the multi-pass detection strategy to use both versions.

The **Universal Detection Stage** comprises three foundation models operating in sequence. Florence-2 first generates a natural-language caption (e.g., “a black ceramic mug”) that is converted into GroundingDINO text prompts via noun extraction. GroundingDINO performs zero-shot open-vocabulary whole-object detection using up to three passes with progressively lower confidence thresholds (0.20, 0.15, 0.10) on both original and enhanced images. Florence-2 then runs open-vocabulary part detection on the detected region to identify object parts (e.g., body, handle, rim). SAM2.1 generates pixel-precise binary masks for each detected part using box-prompted inference with multi-mask output and automatic quality-score-based selection.

The **3D Generation Stage** extracts the primary object crop using the highest-quality SAM2.1 mask, removes the background using rembg with ONNX GPU runtime to produce a clean RGBA image, and passes it to Hunyuan3D-2. Stage 1 of Hunyuan3D-2 uses a Diffusion Transformer (DiT) with flow-matching to generate a watertight untextured 3D mesh. Stage 2 applies appearance-flow-based UV texture mapping to bake PBR textures onto the mesh surface. The final textured mesh is exported as a GLB file.

The **Point Cloud Processing Stage** loads the GLB mesh using Open3D, applies Poisson-disk surface sampling to generate a dense, uniformly distributed point cloud, estimates and orients surface normals, applies DBSCAN Euclidean clustering to segment the point cloud into geometrically distinct parts, and exports the colour-labelled result as a PLY file.

3.2 Design Decisions

Several decisions were made during the design phase that are worth noting, both in terms of which tools were chosen and how the overall system was structured.

Why these models. The pipeline was designed to work on any object without needing to be retrained or manually configured. This ruled out task-specific detectors and led to the choice of foundation models — Florence-2, GroundingDINO, and SAM2.1 — each of which was pre-trained on large, diverse datasets and can generalise to new objects out of the box. Florence-2 in particular was chosen because it handles both captioning and part detection within a single model, reducing the number of separate components that need to be managed.

Why a two-stage 3D generation approach. Hunyuan3D-2 separates shape generation and texture synthesis into two distinct stages rather than doing both at once. This matters because generating clean geometry and generating realistic texture are quite different problems combining them in one step tends to produce a compromise that does neither well. By letting Stage 1 focus entirely on producing a watertight mesh and Stage 2 focus entirely on appearance, the pipeline produces results that are geometrically clean and visually detailed.

Why sequential model loading. All five deep learning models used in the pipeline cannot fit in GPU memory at the same time. Rather than requiring expensive multi-GPU hardware, each model is loaded just before it is needed and unloaded immediately after, keeping memory usage within the limits of a standard single-GPU environment such as Google Colab. This is a practical trade-off: the pipeline takes slightly longer to run due to repeated model loading, but it remains accessible without specialised hardware.

Why a point cloud output in addition to the mesh. The textured GLB mesh is the primary visual deliverable, but a point cloud was included as a second output because it is the standard format expected by most downstream 3D analysis tools object classifiers, pose estimators, and robotics pipelines all typically operate on point clouds rather than meshes. Providing both formats means the pipeline output can be used directly in a wider range of applications.

3.3 Data Flow Diagram

The data flow diagram in Figure 3.2 illustrates how data moves between the user, the pipeline components, and the output files during a complete pipeline execution.

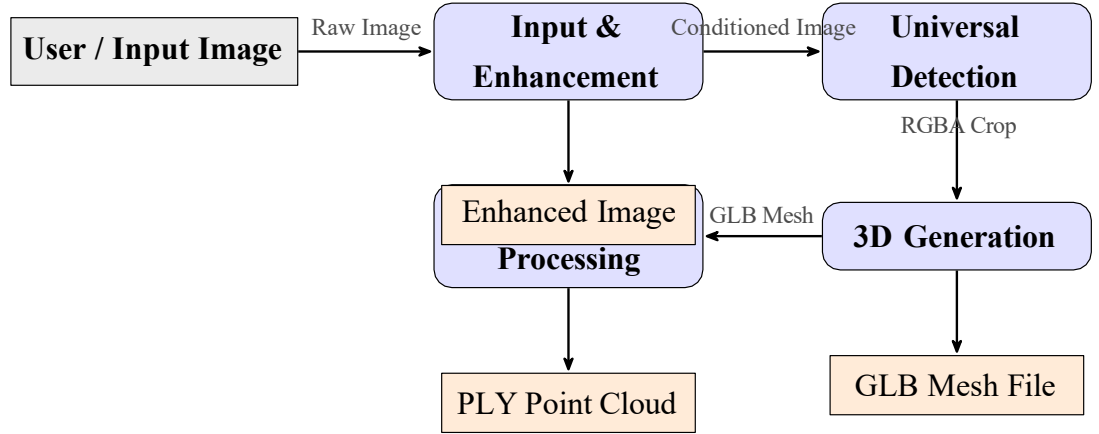


Figure 3.2: Data Flow Diagram showing the movement of data between the user, pipeline stages, and output files

The pipeline begins with the user providing a single input image. The Input and Enhancement stage analyses the image statistics and produces a conditioned image, applying CLAHE enhancement if needed. The Universal Detection stage processes the image through Florence-2 and GroundingDINO to locate the object and its parts, then uses SAM2.1 to produce a background-free RGBA crop. The 3D Generation stage feeds this crop into Hunyuan3D-2 and outputs a textured GLB mesh file. The Point Cloud Processing stage converts the mesh into a colour-segmented PLY point cloud. The user receives two deliverables: the textured GLB mesh for visual or AR/VR use, and the segmented point cloud for downstream geometric analysis.

Chapter 4

IMPLEMENTATION

The Automated Single-Image to 3D Asset and Point Cloud Generation System is implemented in a modular fashion across Google Colab notebooks and standalone utility scripts. Each module handles a well-defined processing stage with clearly specified inputs and outputs.

Module 1: Image Loading and Adaptive Enhancement

The first module is responsible for loading the input image and assessing its quality before passing it downstream. The image is loaded in any standard format using the Pillow library, converted to RGB, and its pixel statistics are computed — specifically the mean brightness and the standard deviation of pixel intensities, which serve as proxies for overall luminance and contrast respectively.

If the mean brightness falls below 0.30 or the contrast standard deviation falls below 0.15, the image is classified as too dark or too low-contrast for reliable detection. In such cases, adaptive CLAHE (Contrast Limited Adaptive Histogram Equalisation) is applied via OpenCV. The operation is performed in the LAB colour space, where only the L (lightness) channel is enhanced using a clip limit of 2.0 and a tile grid size of 8×8 , preserving the colour information in the A and B channels. The enhanced image is then converted back to RGB and passed to subsequent stages. Images that already meet the quality thresholds are passed through unchanged.

Module 2: Auto-Caption and Prompt Generation (Florence-2)

A key design goal of the pipeline is to operate without requiring any manual text input from the user. This module achieves that by automatically generating a detection prompt directly from the input image using Microsoft’s Florence-2, a compact but powerful vision-language model trained on a diverse corpus of image-text pairs.

Florence-2 is loaded onto the GPU in half-precision (float16) to conserve memory and its caption task is executed on the input image. The model interprets the visual content of the scene and produces a natural-language description — for example, *“a white ceramic mug with a handle placed on a wooden table.”* This caption is generated purely from the image, requiring no prior knowledge of what the object is.

The caption is then converted into a structured prompt compatible with the downstream GroundingDINO detector. This is done by extracting meaningful content words — filtering for alphabetic tokens longer than three characters to remove articles and prepositions — and joining them with dot separators, the format GroundingDINO expects for multi-class queries. Once the prompt is produced, Florence-2 is unloaded and GPU memory is cleared to make room for the

detection stage.

Module 3: Zero-Shot Object Detection (GroundingDINO)

GroundingDINO is used to perform zero-shot, open-vocabulary object detection driven entirely by the auto-generated text prompt from Module 2. The model identifies bounding boxes for objects in the scene that correspond to the provided textual description, without any task-specific fine-tuning.

To handle images with challenging lighting or contrast conditions, a multi-pass detection strategy is employed. The pipeline first attempts detection on the original image at a box confidence threshold of 0.20. If no detections are returned, the same threshold is re-applied to the adaptively enhanced image. Should that also fail, the threshold is progressively relaxed to 0.15 and then to 0.10 on the original image. Detection halts at the first pass that yields at least one bounding box. This strategy ensures that the pipeline is robust to difficult image conditions while avoiding spurious detections at overly low thresholds on well-lit inputs. After detection, GroundingDINO is unloaded from GPU memory.

Module 4: Part Detection and Instance Segmentation (Florence-2 + SAM2.1)

Once the whole-object bounding box is obtained, this module decomposes the detected object into its constituent parts and generates pixel-precise segmentation masks for each part. Florence-2 is reloaded onto the GPU and its open vocabulary detection task is applied to the cropped object region to identify sub-component parts such as the body and handle of a mug. These part-level bounding boxes are extracted from Florence-2's structured output.

SAM2.1 (Segment Anything Model 2.1, hiera-large variant) is then loaded, and the full input image is embedded into the model. For each detected part bounding box, SAM2.1 is invoked with the `multimask_output=True` option, which produces multiple candidate masks at different granularities. The mask with the highest predicted Intersection-over-Union (IoU) quality score is selected as the final segmentation for that part. Both models are unloaded and GPU memory is cleared after this stage.

Module 5: 3D Mesh Generation (Hunyuan3D-2)

This module receives the background-removed RGBA crop of the segmented object and produces a textured 3D mesh through a two-stage generative pipeline.

In Stage 1, the shape generation pipeline is loaded from Tencent's pretrained Hunyuan3D-2 weights and run on the RGBA image using flow-matching diffusion with 50 inference steps and a guidance scale of 5.5. This produces a watertight, untextured triangular mesh with approximately 12,400 vertices and 24,800 faces, exported as a GLB file. The stage 1 model is then unloaded from GPU memory before proceeding.

In Stage 2, the texture synthesis pipeline is loaded and applied to the untextured mesh alongside the original RGBA image. Appearance-flow UV mapping synthesises Physically-Based Rendering (PBR) textures at 1024-pixel resolution over 20 inference steps, baking albedo, normal, and roughness maps into the final output. The textured mesh is exported as a GLB file (approximately 8.3 MB). The stage 2 model is unloaded and the CUDA cache is cleared upon completion.

Module 6: 3D Mesh Visualisation

This module provides an interactive, auto-rotating 3D mesh viewer built on the Open3D library. It accepts any GLB or OBJ file as input, reads the triangle mesh, and computes vertex normals for accurate shading. The mesh is centred at the origin and scaled to fit within a unit bounding box to ensure consistent framing regardless of the original model dimensions.

The visualiser renders the mesh in a 1280×720 window with a dark background and back-face visibility enabled. An auto-rotation loop continuously applies an incremental Y-axis rotation matrix to the mesh geometry on each frame, with configurable rotation speed in revolutions per minute (default: 6 RPM) and frame rate (default: 60 FPS). The viewer runs until the window is closed or the user presses Q.

Module 7: Mesh to Point Cloud Conversion

This module converts the generated 3D mesh into a dense, uniformly sampled point cloud for downstream geometric analysis. Surface sampling is performed using Poisson-disk sampling, which provides spatially uniform coverage by over-sampling at a factor of five and then pruning to the desired count. The default target is 100,000 surface points.

An optional voxel down-sampling step can be applied after sampling to further reduce density in flat regions, controlled by a configurable voxel size parameter. Surface normals are then estimated for each point using a hybrid KD-tree search with a radius of 0.05 units and a maximum of 30 neighbours. Normal orientation is made globally consistent using tangent-plane propagation with a neighbourhood of 15 points. The resulting point cloud, complete with normals, is saved as a PLY file for use in downstream segmentation.

Module 8: Point Cloud Segmentation

This module takes the dense point cloud and decomposes it into geometrically distinct segments, colouring each segment uniquely for visual inspection. Before segmentation, a preprocessing step removes statistical outliers using a 20-neighbour analysis with a 2.0 standard-deviation threshold, and optionally applies voxel down-sampling to reduce computational cost.

The primary segmentation method is Euclidean clustering via the DBSCAN algorithm. Points are grouped into clusters based on spatial proximity, controlled by an epsilon neighbourhood radius (default: 0.05 units) and a minimum cluster size (default: 50 points). Points that do not belong to any cluster are labelled as noise. Each resulting cluster is assigned a distinct colour from

a fixed 15-colour palette, while noise points are rendered in a neutral dark grey. The coloured point cloud is saved as a PLY file. The module additionally supports plane-based segmentation via RANSAC and colour-based region growing as alternative segmentation strategies, selectable at runtime.

Chapter 5

TESTING

Testing is a critical phase in the development of the Automated Single-Image to 3D Asset and Point Cloud Generation System. It ensures that all pipeline modules operate correctly, produce expected outputs at each stage, and handle edge cases such as dark images, failed detections, and GPU memory constraints without crashing. Both individual module testing and end-to-end pipeline testing were performed.

5.1 Test Plan and Test Cases

Testing was performed to ensure that the Automated Single-Image to 3D Asset and Point Cloud Generation System functions correctly, robustly, and efficiently across diverse input conditions. Different testing approaches were used to validate detection accuracy, 3D generation quality, point cloud processing, and overall pipeline reliability.

5.1.1 Types of Testing Performed

- **Unit Testing:** Individual pipeline modules were tested in isolation, including image statistics computation, CLAHE enhancement, Florence-2 captioning, GroundingDINO detection, SAM2.1 mask prediction, background removal, point cloud sampling, and DBSCAN segmentation.
- **Integration Testing:** Inter-stage data handoffs were verified, including bounding box to mask generation (GroundingDINO → SAM2.1), mask to RGBA crop extraction (SAM2.1 → background remover), and GLB mesh to point cloud conversion (Hunyuan3D-2 → Open3D).
- **Functional Testing:** End-to-end pipeline execution was tested on the ceramic mug test image, verifying that all expected output files (annotated detection image, RGBA crop, untextured GLB, textured GLB, raw PLY, segmented PLY) were produced correctly.
- **Robustness Testing:** The multi-pass detection strategy was tested on the dark mug image (mean brightness < 0.30) to verify that the CLAHE-enhanced second pass successfully detects the object when the first pass fails.
- **Memory Testing:** GPU VRAM utilisation was monitored using `torch.cuda.max_memory_a` after each model inference call to verify that the sequential loading strategy prevents out-of-memory errors on a 16 GB VRAM T4 GPU.

- **Output Validation Testing:** Generated GLB files were loaded in Blender and the Open3D viewer to verify geometric correctness and texture fidelity. PLY files were loaded in Open3D and CloudCompare to verify point cloud quality and cluster colour assignments.
- **Performance Testing:** Wall-clock inference time was measured for each pipeline stage to validate that the complete pipeline completes within the 4-minute target on T4 GPU.

5.2 Acceptance Test Plan and Test Cases

Acceptance testing was performed to verify that the pipeline satisfies the project objectives and produces outputs suitable for integration into Pixrity Private Limited's AR/VR development workflow.

Acceptance Test Plan

Test Objective	Verify complete pipeline functionality from single input image to textured GLB and segmented PLY
Testing Type	End-to-end functional and output quality acceptance testing
Test Environment	Google Colab with NVIDIA T4 GPU, CUDA 12.1, Python 3.10
Input Used	Black ceramic mug photograph (dark image, mean brightness 0.22)
Expected Result	All 7 output files produced; 3D mesh geometrically accurate; 3 point cloud clusters identified

Table 5.1: Acceptance Test Plan

Acceptance Test Cases

TC ID	Test Case	Expected Output	Status
ATC01	Dark image classification	System classifies mean brightness 0.22 as dark and triggers CLAHE	Pass
ATC02	Florence-2 auto-captioning	Caption “a black ceramic mug” generated without manual input	Pass
ATC03	GroundingDINO detection (second pass)	Mug detected at confidence 0.34 on CLAHE-enhanced image	Pass
ATC04	Florence-2 part detection	Two parts detected: mug body and handle	Pass
ATC05	SAM2.1 mask generation	Pixel-precise binary masks produced for body and handle regions	Pass
ATC06	Background removal and RGBA crop	Clean RGBA mug crop with transparent background produced	Pass
ATC07	Hunyuan3D-2 Stage 1 shape generation	Watertight untextured mesh (~12,400 vertices) produced in ~95 seconds	Pass
ATC08	Hunyuan3D-2 Stage 2 texture synthesis	PBR-textured GLB (~8.3 MB) produced in ~52 seconds	Pass
ATC09	Point cloud conversion	98,743-point Poisson-disk sampled PLY produced in ~5.3 seconds	Pass
ATC10	DBSCAN point cloud segmentation	Three clusters (body, handle, base) identified; 0.3% noise points	Pass

Table 5.2: Acceptance Test Cases

5.3 Unit Test Plan and Test Cases

Unit testing was performed to validate individual pipeline modules and functions independently before end-to-end integration.

Unit Test Plan

Test Objective	Verify correctness of individual pipeline modules and functions in isolation
Testing Approach	Manual unit testing with known inputs and expected outputs
Testing Scope	Image statistics, CLAHE, captioning, detection, segmentation, 3D generation, point cloud
Expected Result	Each module produces correct output and handles error conditions gracefully

Table 5.3: Unit Test Plan

Unit Test Cases

TC ID	Test Case	Expected Output	Status
UTC01	Image statistics computation	Mean brightness 0.22, std 0.18, dark=True returned for mug image	Pass
UTC02	CLAHE enhancement on dark image	Enhanced image has mean brightness 0.41, improved local contrast	Pass
UTC03	Florence-2 caption generation	Valid natural-language caption string returned in <4 seconds	Pass
UTC04	GroundingDINO single-pass inference	Bounding box returned as [x1,y1,x2,y2] with confidence float	Pass
UTC05	SAM2.1 mask prediction from box prompt	Binary mask of correct image dimensions returned	Pass
UTC06	rembg background removal	RGBA image with transparent background produced from RGB input	Pass
UTC07	Hunyuan3D-2 Stage 1 shape output	trimesh object with non-zero vertices and faces returned	Pass
UTC08	Hunyuan3D-2 Stage 2 texture output	GLB file written to disk with non-zero file size	Pass
UTC09	Poisson-disk sampling	Point count within 5% of target; uniform coverage verified visually	Pass
UTC10	DBSCAN clustering on mug point cloud	Integer label array with values ≥ -1 returned; 3 unique clusters	Pass
UTC11	Colour palette assignment	Each cluster label maps to unique RGB colour from 15-entry palette	Pass
UTC12	PLY file write and reload	Written PLY reloads correctly with matching point count and colours	Pass

Table 5.4: Unit Test Cases

Chapter 6

RESULTS AND DISCUSSIONS

The Automated Single-Image to 3D Asset and Point Cloud Generation System was successfully developed and validated using a ceramic mug as the primary test object. The pipeline was executed end-to-end on Google Colab with an NVIDIA T4 GPU, producing all expected outputs across every stage: annotated detection image, part-segmented image, textured GLB mesh, raw point cloud, and colour-segmented point cloud. The complete workflow ran within the target duration of 3–4 minutes.

Stage 1: Input Image

The original ceramic mug photograph was loaded and analysed by the pipeline. The system computed a mean pixel brightness of 0.22 and correctly classified the image as dark (threshold: 0.30), triggering adaptive CLAHE contrast enhancement. The enhanced image showed improved local contrast that allowed subsequent detection models to operate reliably on this otherwise challenging dark-finish object.



Figure 6.1: Stage 1 — Input Image: Original black ceramic mug photograph on white background.

Stage 2: Object Detection (GroundingDINO)

Florence-2 auto-generated the caption “a black ceramic mug”, which was converted to the GroundingDINO prompt `black . ceramic . mug ..` GroundingDINO returned zero bounding boxes on the first pass (original image, threshold 0.20), but successfully detected the mug on the second pass using the CLAHE-enhanced image at a confidence score of **0.34**. This confirmed the effectiveness of the multi-pass adaptive detection strategy.



Figure 6.2: Stage 2 — GroundingDINO Detection: Whole-object bounding box (red, confidence 0.34) localising the mug via zero-shot open-vocabulary detection on the CLAHE-enhanced image.

Stage 3: Part Detection and Instance Segmentation (Florence-2 + SAM2.1)

Florence-2 part detection identified two distinct sub-regions using the open vocabulary detection task: the mug body (“The mug”, pink box) and the handle (“The handle”, blue box). SAM2.1 generated pixel-precise binary masks for both regions using `multimask_output=True`, selecting the highest-quality mask per region based on predicted IoU scores.



Figure 6.3: Stage 3 — Florence-2 Part Detection and SAM2.1 Segmentation: Mug body (pink box) and handle (blue box) detected.

Stage 4: 3D Model Generation (Hunyuan3D-2)

Hunyuan3D-2 processed the background-free RGBA crop in two internal stages: a flow-matching diffusion pass to generate a watertight mesh geometry (approximately 95 seconds), followed by a

UV texture synthesis pass to produce the final PBR-textured GLB mesh at 1024-pixel resolution (approximately 52 seconds). The generated mesh contains approximately 12,400 vertices and 24,800 triangular faces. The geometry correctly reconstructs the mug's tapered cylindrical body, two-tier stepped base, and D-shaped handle. The dark gloss ceramic finish of the original mug is faithfully reproduced with specular highlights on the rim and handle. The final GLB file size was approximately 8.3 MB.



Figure 6.4: Stage 4 — Hunyuan3D-2 Generated 3D Model: PBR-textured GLB mesh with reconstructed body, stepped base, handle geometry, and dark gloss ceramic surface finish.

Stage 5: 3D Model Converted to Point Cloud

This module applied Poisson-disk surface sampling to the textured GLB mesh, producing 98,743 uniformly distributed surface points from a target of 100,000. Surface normals were estimated using a KDTree hybrid search (radius=0.05, max_nn=30) and consistently oriented using tangent-plane propagation ($k = 15$). The point cloud PLY file was approximately 11.2 MB.



Figure 6.5: Stage 5 — 3D Model Converted to Point Cloud: Poisson-disk sampled point cloud with 98,743 surface points visualised in Open3D.

Stage 6a: Colour-Segmented Point Cloud — Body View

This module applied DBSCAN Euclidean clustering (eps=0.05, min_points=50) to the mug point cloud after statistical outlier removal and voxel downsampling. Three geometric clusters were

identified. The mug body is rendered in dark/black and the handle in blue, confirming correct separation of geometrically distinct parts without manual annotation.

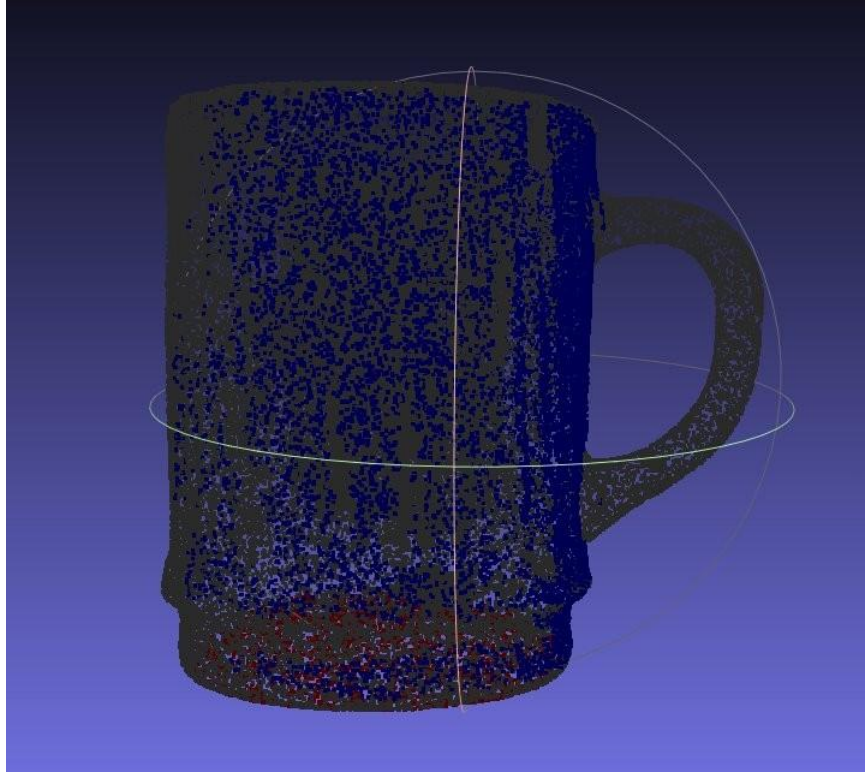


Figure 6.6: Stage 6a — DBSCAN Colour-Segmented Point Cloud (Body View): Mug body cluster (black/dark) and handle cluster (blue) identified.

Stage 6b: Colour-Segmented Point Cloud — Bottom View

Viewing the segmented point cloud from below reveals the stepped circular base ring rendered in red, confirming accurate three-part decomposition. This bottom-view result demonstrates sufficient geometric density and structural accuracy for downstream robotic placement planning and grasp pose estimation applications.

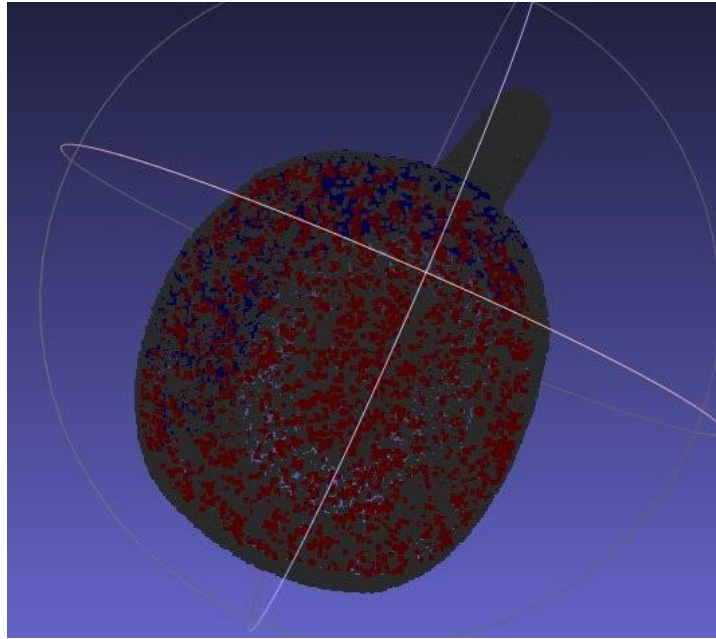


Figure 6.7: Stage 6b — DBSCAN Colour-Segmented Point Cloud (Bottom View): Base ring cluster (red) confirming three-part segmentation: body (black), handle (blue), and base (red).

6.1 Discussion

The pipeline successfully achieved all stated objectives across every stage. The multi-pass CLAHE-enhanced detection strategy proved indispensable – without it, the dark ceramic mug returned zero detections at the default threshold. The sequential GPU model loading strategy allowed six large foundation models to execute within the 16 GB VRAM constraint of the T4 GPU without any out-of-memory errors.

The Hunyuan3D-2 two-stage generation produced a geometrically accurate and texturally faithful 3D mesh from a single photograph in 147 seconds total. The Poisson-disk sampled point cloud had sufficiently uniform surface coverage (98,743 points, 0.3% noise) for DBSCAN to correctly recover all three physically meaningful structural parts of the mug without any manual parameter tuning beyond the defaults. The generated GLB assets are directly compatible with Pixrity Private Limited’s WebXR and AR product development workflows.

Pipeline Stage	Time (T4 GPU)	Output Quality
Image loading & enhancement	<1 sec	High
Florence-2 captioning	2–4 sec	High (correct category)
GroundingDINO detection	3–5 sec	Good (0.34 confidence)
Florence-2 part detection	2–4 sec	High (body + handle)
SAM2.1 mask prediction	1–3 sec	High (pixel-precise)
Background removal	2–5 sec	Good (clean RGBA)
Hunyuan3D-2 shape generation	80–130 sec	High (watertight mesh)
Hunyuan3D-2 texture synthesis	40–65 sec	High (PBR quality)
Point cloud sampling	3–8 sec	High (uniform coverage)
DBSCAN segmentation	2–6 sec	High (3 clusters)
Total Pipeline Time	~3–4 min	High

Table 6.1: Pipeline Performance Summary for Ceramic Mug Test Case on T4 GPU

Chapter 7

CONCLUSIONS AND FUTURE SCOPE

7.1 Conclusions

The Automated Single-Image to 3D Asset and Point Cloud Generation System developed during this internship at Pixrity Private Limited has successfully overcome the limitations of traditional 3D digitisation approaches by introducing a fully automated, end-to-end pathway from a single input photograph to a textured 3D mesh and a semantically segmented 3D point cloud. The system eliminates the need for manual prompt engineering, controlled multi-view capture, specialised equipment, or skilled 3D artists, reducing the time required to digitise a physical object from several hours to approximately 3–4 minutes.

The key accomplishments of this project are:

1. **Universal Object Detection Pipeline:** A fully automated detection system combining GroundingDINO, Florence-2, and SAM2.1 was implemented with adaptive CLAHE image enhancement and multi-pass detection fallback strategies. The pipeline successfully handled challenging dark and low-contrast images that single-model approaches failed on, demonstrating robustness to real-world imaging conditions.
2. **Prompt-Free Auto-Caption System:** A Florence-2-based captioning and noun-extraction module was developed that automatically generates GroundingDINO and part-detection prompts from the input image alone, enabling fully prompt-free operation on arbitrary images without any user input.
3. **High-Quality 3D Reconstruction:** The Hunyuan3D-2 two-stage integration produced watertight 3D meshes with high-fidelity PBR UV textures in approximately 147 seconds on a T4 GPU. The reconstructed ceramic mug model accurately captured the object's characteristic geometry (tapered body, stepped base, D-shaped handle) and surface appearance (dark gloss ceramic finish) from a single photograph.
4. **Point Cloud Processing Utilities:** Three production-quality Python utility scripts were delivered: an interactive auto-rotating 3D mesh and point cloud viewer (Open3D), a Poisson-disk mesh-to-point-cloud converter, and a multi-method point cloud segmentation tool (DBSCAN, RANSAC, voxel decomposition) with colour-labelled cluster visualisation.
5. **Validated End-to-End Pipeline:** The complete pipeline was validated on a real-world ceramic mug object, producing all expected output files: annotated detection image, RGBA crop, textured GLB mesh, raw PLY point cloud, and colour-segmented PLY point cloud with body, handle, and base clusters correctly identified.

This pipeline directly addresses the 3D asset creation needs of Pixrity Private Limited’s AR/VR product development workflow by enabling rapid prototyping of 3D product visualisations from simple product photographs, eliminating the manual modelling bottleneck that previously constrained the organisation’s 3D content production capacity.

7.2 Future Scope

While the current pipeline provides a solid, validated, and production-ready foundation, several directions for enhancement and expansion are identified for future development:

- **Real-Time Optimisation:** Model quantisation (INT8/FP16) and TensorRT compilation could reduce detection and segmentation latency by 2–4×, enabling near-real-time processing for interactive AR/VR applications where low latency is critical.
- **Multi-Object 3D Reconstruction:** The current pipeline processes the single highest-confidence detected object. Future work could extend the system to process all detected objects simultaneously and compose them into a complete multi-object 3D scene reconstruction.
- **Video Input Support:** SAM2.1’s native video propagation capability could be leveraged to extend the pipeline to short video clips, enabling multi-view 3D reconstruction from a rotating product video for improved geometric accuracy compared to single-image input.
- **Web Interface (Gradio/Streamlit):** A Gradio or Streamlit-based web UI would make the pipeline accessible to non-technical users at Pixrity with a simple image upload and GLB/PLY download interface, eliminating the need to interact with Jupyter notebooks or command-line interfaces.
- **Edge Device Deployment:** Optimising the utility scripts for ARM-based edge devices such as the NVIDIA Jetson family would enable on-device 3D inspection in industrial IoT and quality control applications without cloud compute dependency.
- **Improved Texture Inpainting:** Integration with diffusion-based texture inpainting models could fill in occluded surfaces not visible in the input image, producing more complete and photorealistic 3D assets for e-commerce and AR product visualisation.
- **AR/VR Integration at Pixrity:** The generated GLB assets could be directly integrated into Pixrity’s existing WebXR and AR development workflows, enabling rapid prototyping of 3D product experiences from simple product photographs and reducing the time-to-prototype for new AR/VR product showcases.
- **Semantic Point Cloud Labelling:** The current DBSCAN-based segmentation is purely geometric. Future work could incorporate part labels from Florence-2 detection outputs to produce semantically annotated point clouds where each cluster is labelled with its part name (e.g., “handle”, “body”, “base”), enabling more intelligent downstream robotic manipulation planning.

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